

Experiment No. 15

Exploring Emerging Business Intelligence Technologies and Their Applications in Future Retail

Business Intelligence & Analytics Laboratory (E1CSC25703) – Alliance University

1. Aim

To investigate emerging Business Intelligence technologies – AI-driven analytics, cloud BI, real-time BI and advanced visualization – and demonstrate their potential applications in a future retail business scenario.

2. Objectives

- Clean and prepare transactional retail data for analysis.
- Build traditional BI measures and dashboards (KPIs, trend, country, product).
- Apply AI-assisted analytics: customer segmentation (RFM + K-Means), key influencers, anomaly detection and forecasting.
- Demonstrate real-time BI through a simulated live sales feed.
- Describe cloud BI publishing and interactive/advanced visualization.
- Compare technologies and discuss qualitative ROI.

3. Dataset

UCI Online Retail dataset: 541,909 transactions of a UK-based online retailer (1 Dec 2010 – 9 Dec 2011) with InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. The last month is partial (data ends 9 Dec 2011).

4. Software / Tools

Power BI Desktop and Power BI Service (dashboard, publishing); Python (pandas, scikit-learn, statsmodels) for AI analytics; Microsoft Excel for result tables; HTML/JavaScript for the real-time simulation.

5. Data Preprocessing

Check	Records
Rows loaded	541,909
Missing CustomerID	135,080
Missing Description	1,454
Cancelled invoices (C...)	9,288
Quantity <= 0	10,624
UnitPrice <= 0	2,517
Duplicate rows	5,268
Rows after cleaning (sales analysis)	522,713
Rows with CustomerID (customer analysis)	391,286

Cancelled invoices (starting with C), non-positive quantity/price, duplicates and non-product stock codes (postage, fees, manual entries) were removed. Records without CustomerID were kept for sales analysis but excluded from customer-level (RFM) analysis. SalesAmount = Quantity × UnitPrice.

6. Emerging BI Technologies

AI-driven analytics: Machine-learning methods find patterns automatically: segmentation, drivers, anomalies and forecasts.

Cloud BI: Reports are published to Power BI Service for sharing, scheduled refresh and access from anywhere.

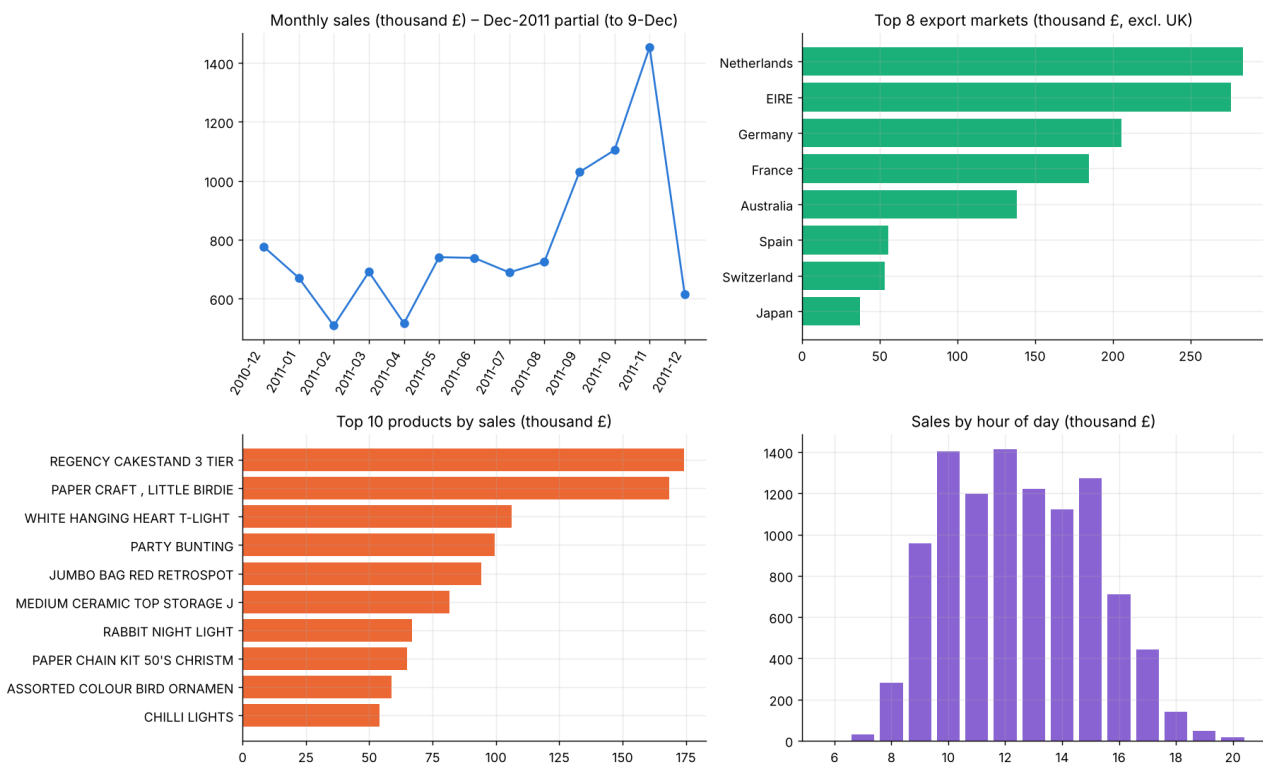
Real-time BI: Continuous ingestion of transactions so managers see current sales; here simulated with a streaming-style feed.

Advanced visualization: Interactive dashboards with drill-down, maps, treemaps, heat maps and cross-filtering.

7. Implementation and Results

7.1 Traditional BI – KPIs

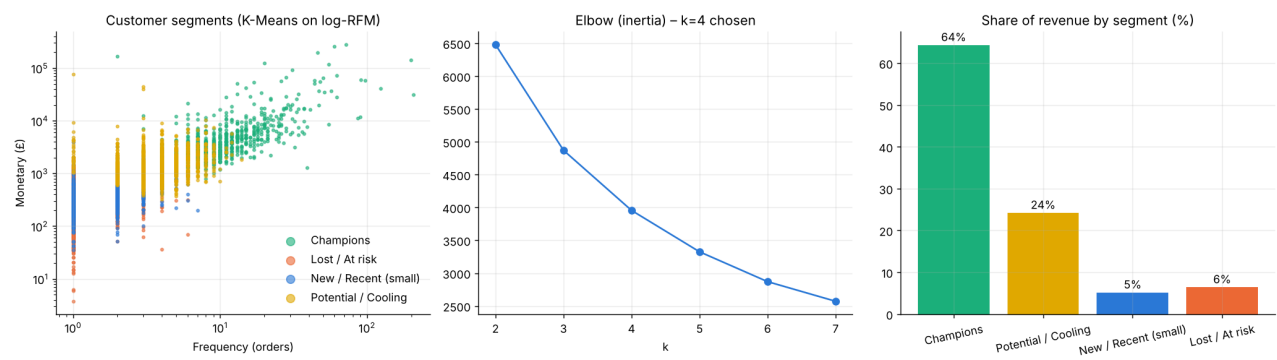
KPI	Value
Total Sales	£10,254,959
Total Orders	19,776
Total Customers	4,334
Total Quantity	5,561,563
Average Order Value	£518.56



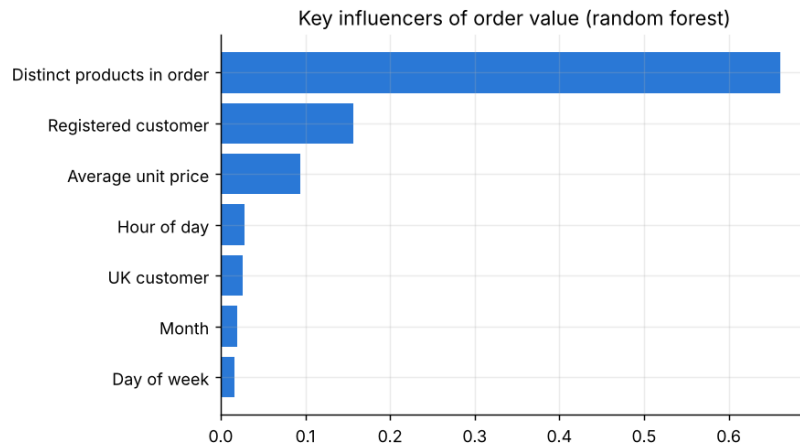
7.2 AI-driven analytics – customer segmentation (RFM + K-Means, k = 4)

Segment	Customers	Avg recency (days)	Avg orders	Avg spend £	Revenue %
Champions	691	11	13.9	8,136	64.3
Potential / Cooling	1188	70	4.1	1,780	24.2
New / Recent (small)	813	17	2.2	552	5.1

Segment	Customers	Avg recency (days)	Avg orders	Avg spend £	Revenue %
Lost / At risk	1642	180	1.3	341	6.4

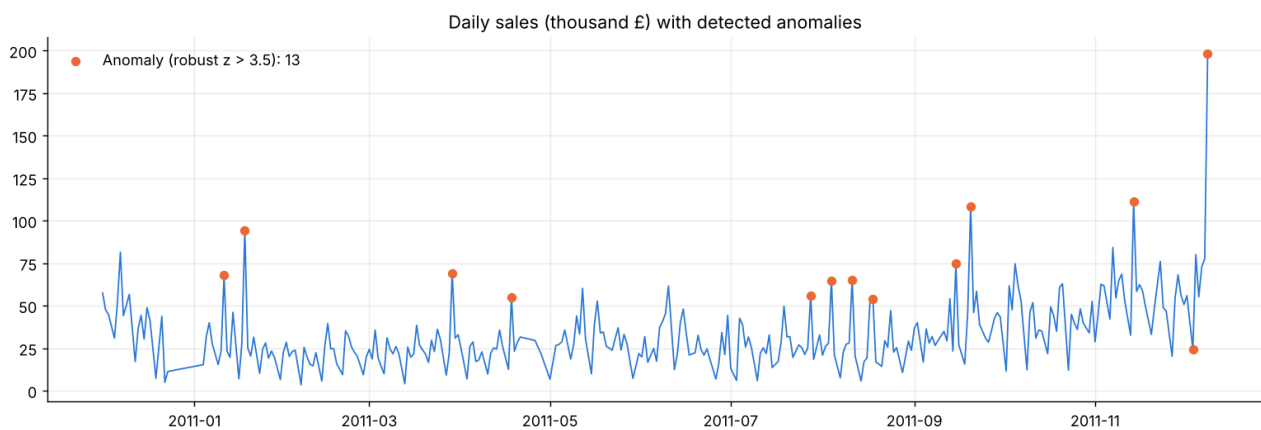


7.3 Key influencers of order value



A random-forest model on invoice-level data shows the number of distinct products per order is the dominant driver of order value, followed by registered-customer status and average unit price. Country, hour and weekday matter little.

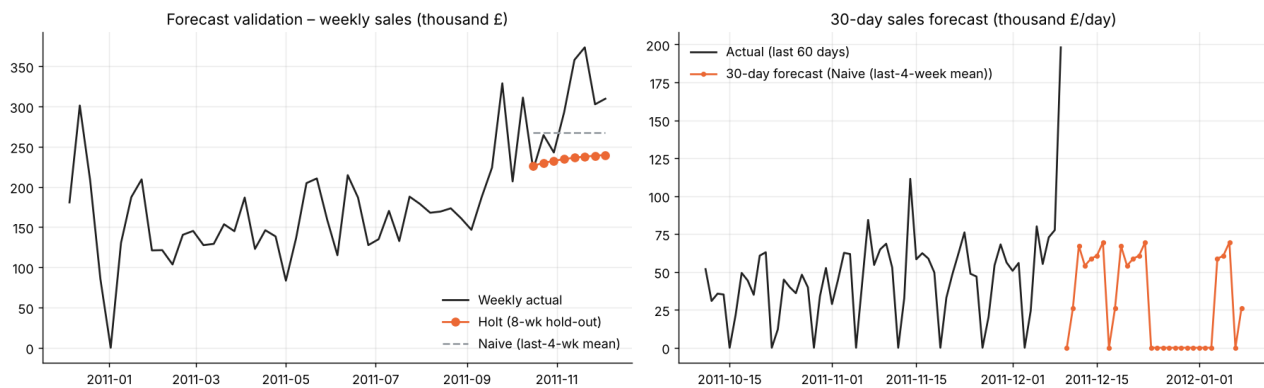
7.4 Anomaly detection



Days whose sales deviate more than 3.5 robust standard deviations from a rolling median were flagged: 13 days (12 spikes, 1 drop) – candidates for bulk orders, promotions or data issues.

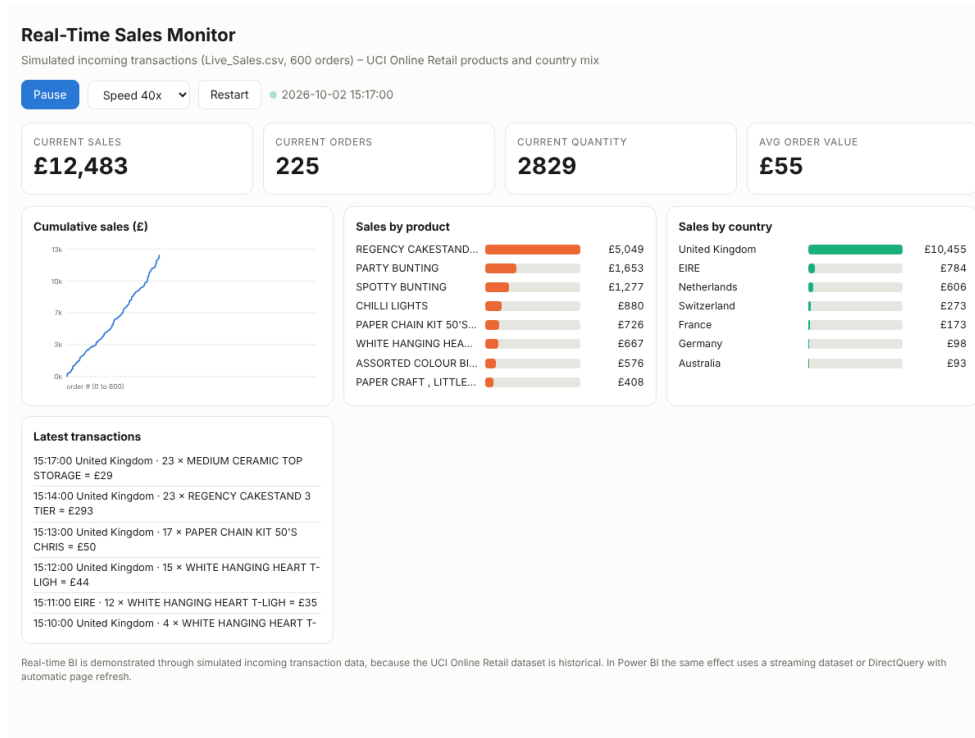
7.5 Forecasting

Model	MAE (£/week)	MAPE %
Holt damped trend	62,415	19.2
Simple exp. smoothing	61,053	18.8
Naive (last-4-week mean)	46,571	14.9



On the 8-week hold-out (the pre-Christmas surge), trend models under-predicted and the naive 4-week mean performed best. With only one year of data, seasonality cannot be learnt, so the 30-day forecast is indicative; the Christmas shutdown (24 Dec – 3 Jan, seen in the data) was applied.

7.6 Real-time BI (simulation)



Real-time BI was demonstrated through simulated incoming transaction data because the UCI Online Retail dataset is a historical transactional dataset. The 600-row Live_Sales.csv feed streams into a monitor showing current sales, orders, quantity, sales by product and by country.

7.7 Cloud BI and advanced visualization

The Power BI report (3 pages: BI Overview, AI-Driven Analytics, Advanced Visualization) is published from Power BI Desktop with Home → Publish to a workspace; key visuals are pinned to a cloud dashboard, shared with authorized stakeholders and refreshed on a schedule. Slicers (Country, Year, Month, Product) cross-filter all visuals; map, treemap and heat-map matrix provide advanced visualization.

8. Future Business Scenario – AI-Powered Smart Retail BI System



The retailer continuously monitors sales, predicts demand, identifies high-performing products, detects unusual sales patterns, offers interactive dashboards and supports management decisions through the workflow above.

9. Comparison of Emerging Technologies

Technology	Application	Scalability	Implementation effort	Business benefit
AI-driven Analytics	Customer segmentation, anomaly detection, forecasting	High	Moderate	Automated insights
Cloud BI	Online reporting, sharing, scheduled refresh	High	Moderate	Anywhere access

Technology	Application	Scalability	Implementation effort	Business benefit
Real-Time BI	Continuous monitoring of live sales	High	High	Faster response
Advanced Visualization	Interactive dashboards, map, drill-down	High	Low–Moderate	Better understanding

10. Qualitative ROI Analysis

Factor	Traditional BI	Emerging BI
Data analysis	Manual / interactive	AI-assisted (RFM, anomalies, key influencers)
Reporting	Static / interactive	Interactive + automated refresh
Forecasting	Separate analysis	Integrated in dashboard
Monitoring	Periodic	Near-real-time architecture
Collaboration	Local files	Cloud-based sharing
Decision support	Historical	Historical + predictive

No numerical ROI is claimed because actual implementation costs and benefits are not available.

11. Results and Discussion – Key Insights

1. UK = 85.1% of sales; top export markets: Netherlands, EIRE, Germany.
2. Peak month 2011-11 (£1,453,266); Nov is highest, Q4 seasonal build-up before Christmas.
3. Top product: REGENCY CAKESTAND 3 TIER (£174,157).
4. Champions segment = 64% of revenue from 691 customers (16% of base).
5. 13 anomalous days detected (spikes/drops) – candidates for stock/promo/outage review.
6. 30-day forecast $\approx$ £886,729 using Naive (last-4-week mean) (hold-out MAPE 14.9%); Christmas shutdown (24-Dec to 3-Jan) applied; only 1 year of history, so seasonality is not learnt – treat as indicative.

12. Conclusion

The experiment showed how emerging BI technologies extend conventional reporting into AI-assisted analytics, predictive analysis, cloud sharing, near-real-time monitoring and interactive visualization, giving retail managers a more proactive decision-support capability. Limitations: one year of history, partial last month and simulated real-time data.